

Ashlin Varughese Thomas

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Education

National Institute of Science Education and Research (NISER) <i>Integrated MSc. (Major: Physics, Minor: Mathematics) CGPA: 9.79/10.0</i>	Bhubaneswar, India 2022 – Present
Currently enrolled in the penultimate semester; achieved the highest CGPA in the batch for academic years 2022-23, 2023-24, 2024-25, and 2025-26.	
Citadel Residential School <i>AISSCE (Grade 12) 99.2% (District Topper, Centum in Physics & Chemistry)</i>	Kerala, India 2021 – 2022

Research Interests

Core Areas: Condensed Matter Physics, Open Quantum Systems, Quantum Many-Body Physics, Non-Hermitian Physics and Topology, Mathematical Physics, Quantum Information, Nonlinear Dynamics.

Research Experience

Centre for Quantum Dynamics, Griffith University <i>Research Intern Advisor: Prof. Dr. Howard Wiseman & Dr. Lucas Ostrowski</i>	Australia Summer 2026
- Developing a full quantum model of a cascaded amplifier system for circuit QED setups using superconducting circuits to surpass the standard quantum limit of laser coherence. - Analyzing the coherence properties of output beams in the presence of losses, Kerr nonlinearity, and experimental imperfections.	
Quantum and Complex Systems Group, Stockholm University <i>Research Intern Advisor: Prof. Dr. Emil J. Bergholtz & Dr. Paolo Mollinari</i>	Sweden Winter 2026
- Investigated the phase diagram of the extended Bose-Hubbard model in the presence of non-reciprocal hopping. - Executed numerical simulations utilizing exact diagonalization via the ‘edith’ package to analyze the interplay between the non-Hermitian skin effect and novel phase transitions.	
Max Planck Institute for the Physics of Complex Systems <i>Research Intern Advisor: Prof. Dr. Jan Michael Rost & Prof. Dr. Ulf Saalmann</i>	Dresden, Germany Summer 2025
- Engineered a unitary transformation framework to convert quantum Hamiltonians into ‘smooth images’, creating optimized matrix representations for machine learning applications. - Solved underlying non-convex optimization problems on the unitary manifold using Lagrange multipliers and analyzed structural robustness against noise.	
Raman Research Institute (RRI) <i>Research Intern Advisor: Dr. Shovan Dutta (Collab: Prof. M. Haque & Dr. F. Fritzsche)</i>	Bangalore, India Summer 2024
- Formulated a semiclassical quantization scheme for Lindblad dynamics in open quantum systems. - Proposed a quasiprobability measure constructed from left and right eigenmodes of the non-Hermitian Liouvillian superoperator to interpret them as a set of coherences.	
IISER Pune <i>Research Intern Advisor: Dr. Anisa Chorwadwala</i>	Pune, India Winter 2023
Shape Optimization: Studied isoperimetric problems and geometric functionals, focusing on configurations that minimize or maximize specific parameters like perimeter or volume under constraints.	
National Institute of Science Education and Research (NISER) <i>Research Intern Advisors: Dr. Sayantani Bhattacharyya & Dr. Anupam Pal Choudhury</i>	Bhubaneswar, India Summer 2023
Nonlinear Dynamics: Modeled population interactions as a set of coupled differential equations and analyzed them using phase-plane analysis and numerical simulations based on nonlinear dynamical systems. Planar Systems of Differential Equations: Investigated the local and global behavior of autonomous planar systems of ordinary differential equations, exploring fixed points, linearization, and stability criteria.	

Publications

Ashlin V Thomas, Felix Fritzsche, Masudul Haque, Shovan Dutta. “**Extracting the physical content of Liouvillian eigenmodes: Semiclassical quantization.**” *arXiv preprint arXiv:2606.20271* (2026). [\[Link\]](#)

Selected Projects & Talks

Turning Quantum Hamiltonians into Images | Talk ([Report](#), [Slides](#))

Vikiran Physics Club

- Presented a mathematical framework for transforming finite-dimensional Hamiltonians into noise-resilient “smooth images” via unitary transformations, based on research conducted at MPIPES Dresden.
- Explored applications of the methodology in visualizing quantum systems and contributing to quantum error mitigation and control protocols.

Harmonizing Classical and Quantum Mechanics | Talk ([Slides](#))

Vikiran Physics Club

- Presented a robust semi-classical approach to quantum mechanics utilizing the Wigner-Weyl transform to bridge the formalisms of classical phase space and quantum Hilbert spaces.

Spherical Model of a Ferromagnet | Project ([Report](#))

- Investigated the spherical model of a ferromagnet as an analytically solvable alternative to the Ising model to explore continuous spin values, critical phenomena, and phase transitions in statistical mechanics.
- Derived thermodynamic properties including free energy, magnetization, and specific heat via saddle-point approximations, and analyzed system behaviors under applied external magnetic fields.

Magnon Bound States in 1D Heisenberg Chain | Project ([Report](#))

- Investigated two-particle bound states using the Jordan-Wigner transformation and two-particle Green’s functions, identifying exact criteria for localized “doublon” emergence.

Diagonalization of U(1) Covariant Lindbladians | Project ([Report](#))

- Developed an exact combinatorial method using weighted directed graphs to find the eigensystem of Lindbladians covariant under U(1) symmetry.

Nonlinear Dynamics | Project ([Report](#))

- Analyzed fixed points, stability, and bifurcations in linear and nonlinear systems utilizing numerical solving methods and phase-plane geometry.
- Constructed and evaluated a second-order nonlinear mathematical model to study the dynamics of interacting populations.

Google PageRank Algorithm | Project ([Report](#))

- Explored the mathematical foundations of network search, modeling the internet using stochastic matrices and resolving computational challenges in algorithmic convergence.

Honors & Awards

INSPIRE Scholarship | *Department of Science and Technology, Govt. of India*

2022 – Present

- Awarded to the top 1% of students nationwide for pursuing advanced studies in natural sciences.

Academic Excellence Award | *NISER Bhubaneswar*

2023 – 2026

- Achieved the highest CGPA among all Int. MSc students for four consecutive academic years.

Teaching & Leadership Experience

Physics and Mathematics Tutor

Kerala, India

Crispr Learning Pvt. Ltd.

2025 – 2026

- Tutored advanced high school students for premier Indian research institute entrance exams, delivering weekly lectures on mechanics, electromagnetism, and calculus.

Project Group Supervisor

NISER Bhubaneswar

Vikiran (Physics Club)

2023 – 2024

- Led a team of students in applying concepts of dynamical systems; instructed weekly sessions on phase space, stability analysis, and numerical methods.

Technical Skills

Languages & Tools: Python, Mathematica, LaTeX, Linux/Bash

Spoken Languages: English (Fluent), Malayalam (Native), Hindi, Tamil